



ASSESSMENT DECLARATION COVER SHEET

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STUDENT (REGISTRATION) NUMBER: 40735842

MODULE TITLE: Advanced Web Technologies

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Eco Habit Tracker: Design Report (Part 1)

Introduction and Project Description

The Eco Habit Tracker is an open-source, community-supported online application that was developed using the Flask web platform. The goal is to make leading a sustainable lifestyle easier and more motivating for individuals. The program helps users create and sustain environmentally friendly behaviours by providing easy tracking and community assistance. It seeks to establish behaviours that preserve regional ecosystems, cut down on energy use, and lessen the usage of plastic. We think that taking action to protect the environment should be a communal as well as a personal duty.

Many people have the desire to live more sustainably, but they are either unsure of where to begin or lack the energy to make the effort themselves. The environmental uses that are available much too frequently overwhelm or scare beginning users. These problems are addressed by this idea, which turns green action into a celebration for groups by making it an approachable and motivating way to advance.

Background Investigation and Evaluation

Analysis of Existing Applications

Although EcoHero provides educational content, it is an individual learning tool instead of a community-based platform and lacks robust habit-tracking capabilities. Although the site has educational value, it lacks the social support necessary to solidify habits over time.

Habitica, on the other hand, offers good gamification features, but its fantasy game aspect will probably overshadow valuable environmental lessons. Light users who just wish to socialize will be deterred by the social aspect, which makes users participate in complicated organizational structures referred to as guilds.

From this study, there is promise in creating an eco-habit tracker. With this tracker, habit tracking would be simple with the assistance of encouraging community members. The most ideal approach would be prioritizing environmental behaviours and also providing social support to foster long-term behaviour.

Annotated Feature List

Feature	What It Does	Why It's Useful
User Accounts	They can make a password and register with an email address.	Members need a private area where they can safely track their progress.
Personal Dashboard	It reflects users activities, their environmental impact, and the duration they adhered to.	It is encouraging to see an improvement so visibly.
Habit Management	Users can add favorite green habits or create their own.	Everyone gauges their life in a different way.
Community Feed	It reflects all members' overall achievements.	When individuals see how far the group has come, they might work even harder as individuals.
Eco-Badges	There are advantages to attaining goals.	Small gains are included in the process of accomplishment.
Group Challenges	There is a timeline in which to plan goals for the group.	Individuals are more likely to come up with good results when working together than when working individually to achieve them.

Site Navigation and Structure

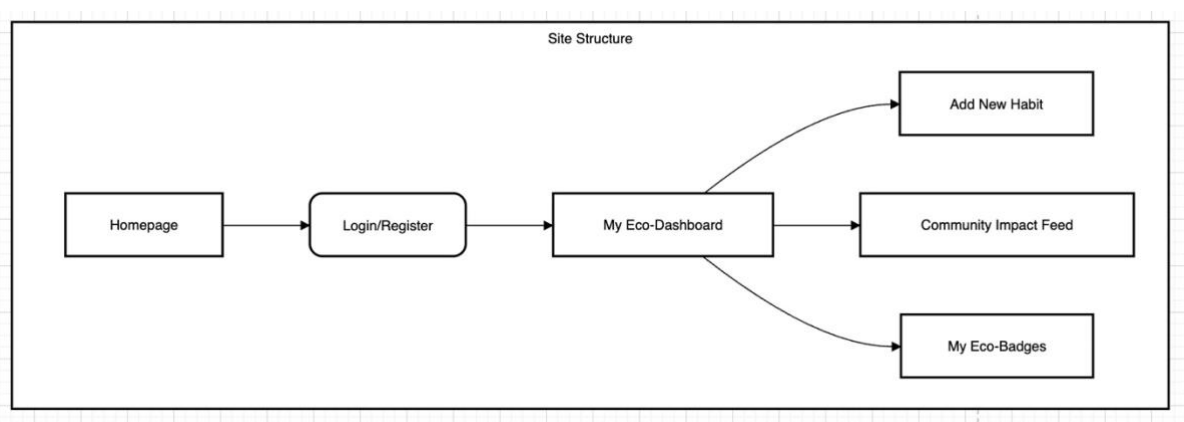


Figure 1: Site Navigation Structure

The straightforward hierarchy of the navigation system does not add any sophistication where users must consider environmental targets. The Eco-Dashboard is front and center in the application. You can access all its major features instantly with a single click. My Eco-Badges, Add New Habit, and

Community Impact Feed are readily available from the dashboard. Its ease of use reduces cognitive strain and directs attention away from difficult navigation and toward habit formation.

Users may access essential services without additional steps or confusion thanks to the straightforward hierarchy of the home page to login/registration and finally to the central dashboard.

User Interface Wireframes and Commentary

Landing Page:

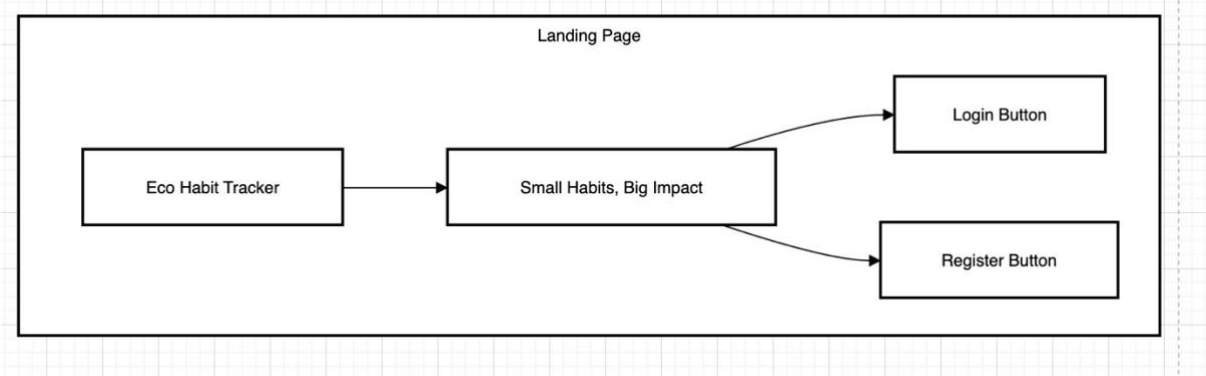


Figure 2: Landing Page Wireframe

Dashboard:

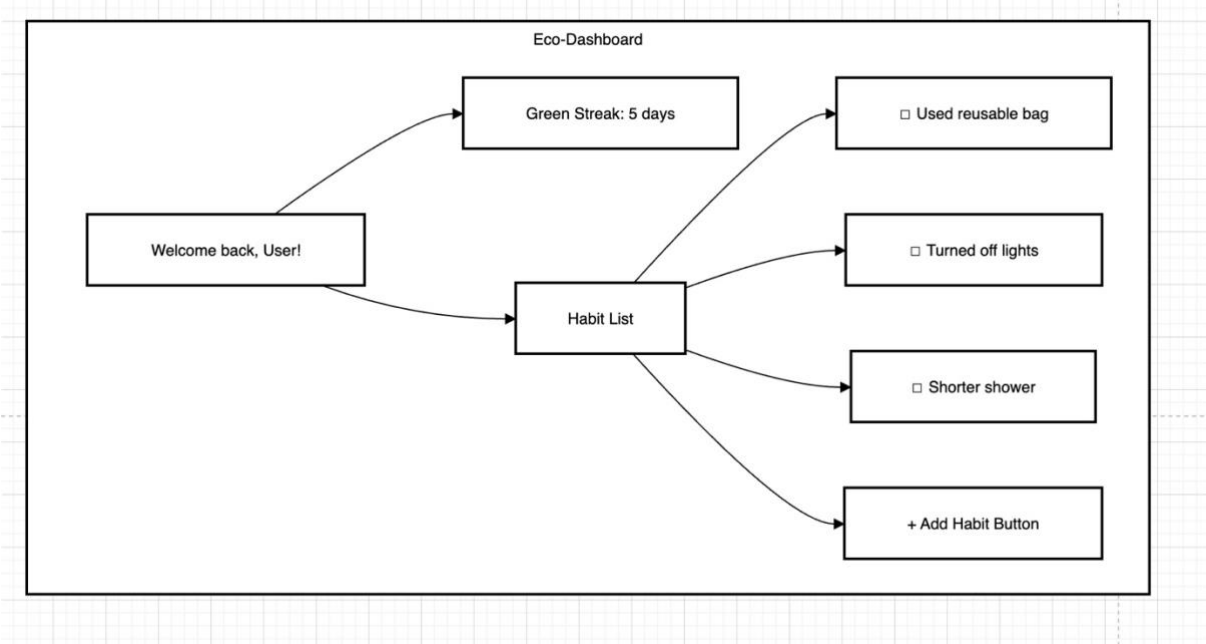


Figure 3: Eco Dashboard Wireframe

Community Feed:

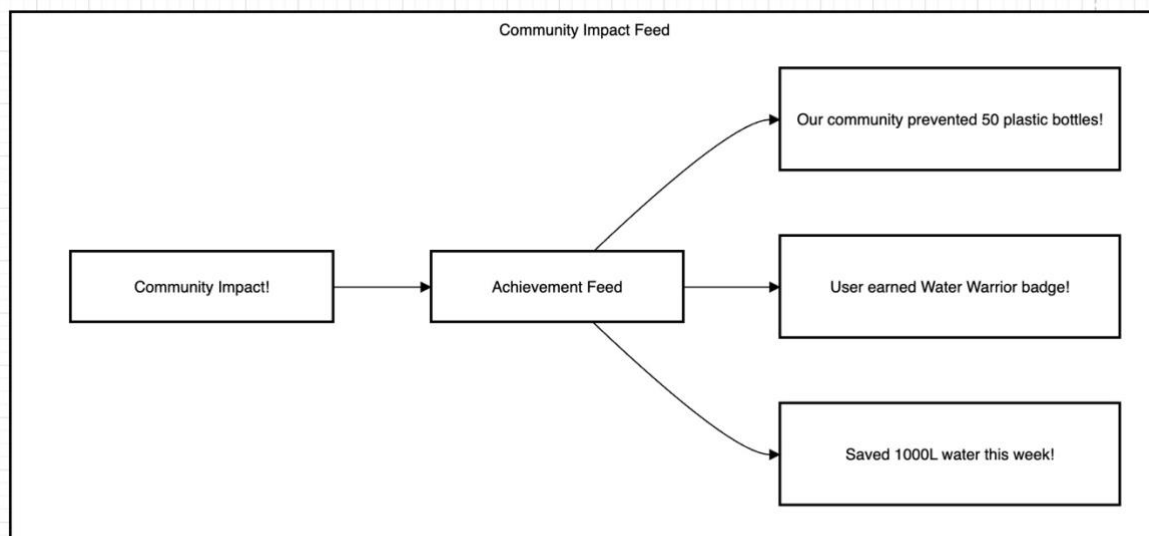


Figure 4: Community Feed Wireframe

Badges Page:

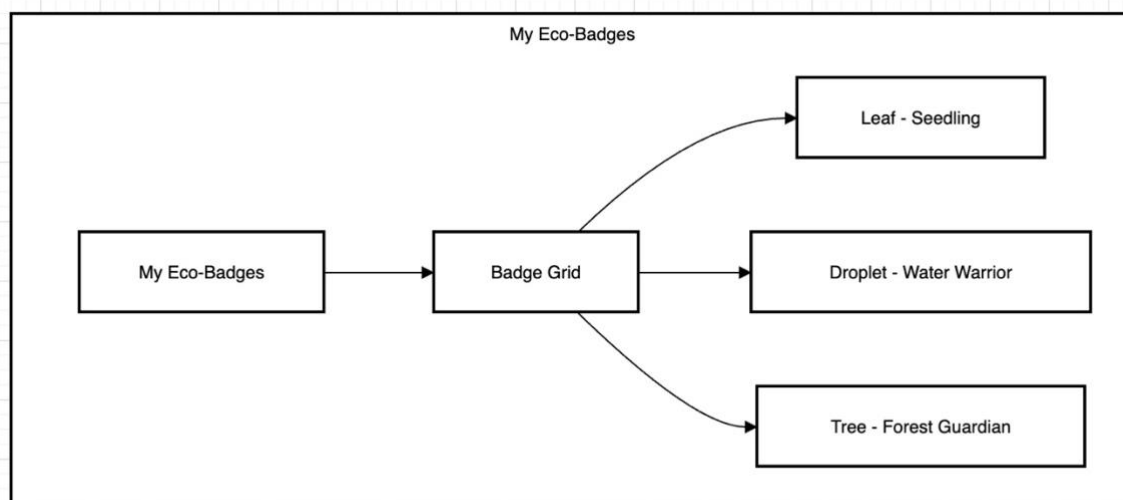


Figure 5: Eco Badge Page Wireframe

The interface design has a green theme but doesn't compromise on style thanks to its subdued colour schemes. The goal of the app is explained by the absence of registration and login icons on the home screen. The primary interface is the Eco-Dashboard. There are checkboxes to record daily activities, a simple green streak count to motivate, and very few possibilities to form new habits. The Community Impact Feed creates social incentive without compromising privacy by sharing anonymous group accomplishments. The Eco-Badges site graphically depicts user progress using a grid of well-known nature symbols.

The interfaces feature a balanced space and a simple information arrangement, making them usable by anyone ranging in age from 16 to 80. The checklist structure has a minimal cognitive burden and promotes high-level adoption of sustainable behaviour.

Implementation Plan

Initial weeks: Full user login, rudimentary Eco-Dashboard setup with habit-based features, and rudimentary Flask with SQLite configuration. Develop a streak calculator algorithm.

Phase in between: Create community elements such as an eco-badge system and an anonymous accomplishment stream. Describe the ecological impact.

The last step is to include a CSS framework into the UI. Prior to launch, testing and problem fixes are made.

Fallback Plan: Begin by using a feature-rich personal habits tracker. After achieving the bare minimum of functioning, add community features only when necessary.

Data Design

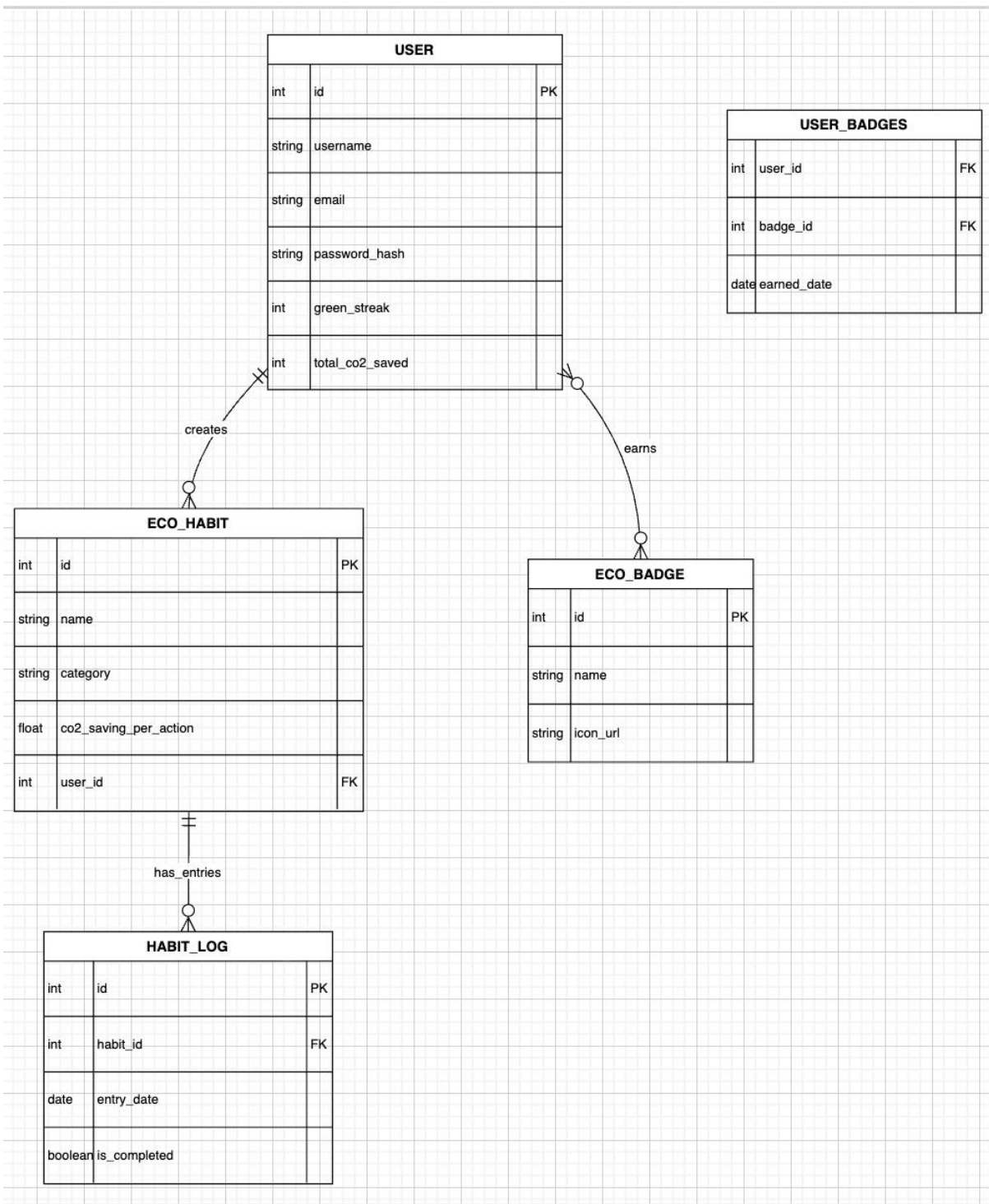


Figure 6: Entity Relationship Diagram

The database schema supports the Eco Habit Tracker app's primary functionality. The USER database contains user progress measures, such as CO₂ saved and green streaks, in addition to user authentication. ECO_HABIT keeps track of habit definitions and calculates the CO₂ impact of each habit using the `co2_saving_per_action` column. With the HABIT_LOG database, users may monitor the fulfilment of their daily routines and help with streak computation. The ECO_BADGE and USER_BADGES databases, which manage the production of Eco Badges and the monitoring of the accomplishments of users who own them, are used to spread the gamification system. Data integrity is ensured by this well-structured database to facilitate the orderly supply of user data.

Additionally, it provides quick and simple access to user dashboards, community features, and development analytics. Table connections preserve the referential integrity of the user's data while enabling comprehensive reporting on the impact on the community and personal growth.

References

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